

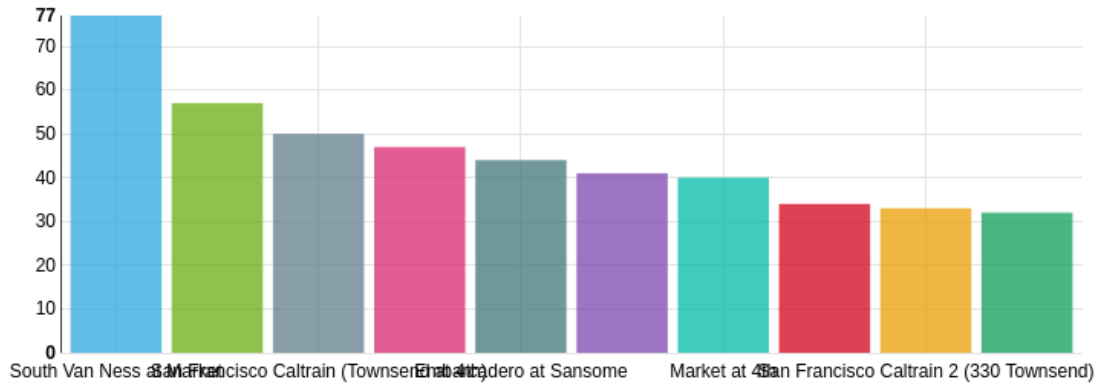
LAB 3: Data Visualization

1. Ten most popular start station.

- a. Business question the visualization attempts to answer.

Which station is the most popular start station. Its information can be used to strategically expand the bike business.

- b. Visualization



- c. Answer obtain (or not) from visualization

Yes, as we can see South Van Ness as Market is the most common starting point for bikers.

- d. Your discussion

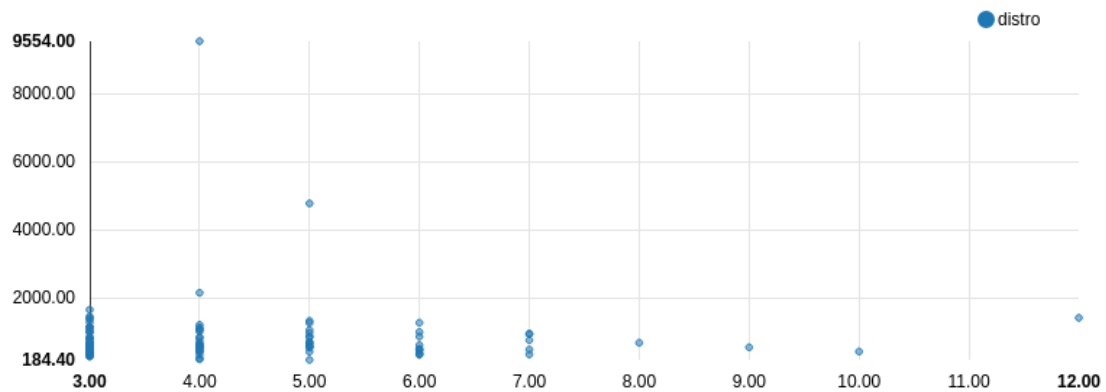
From this bar chart, we can assume the following. First South Van Ness as market came in as the most common spot to start. Followed by Harry Bridges Plaza and San Francisco Caltrain at 57 and 50 bikers respectively.

2. Use Hue or Tableau, demonstrate relationship between the duration of trip and the popularity of such route.

- a. Business question the visualization attempts to answer.

Is there a relationship between the average duration of a trip and the popularity of the route?

b. Visualization



c. Answer obtain (or not) from visualization

No, from this visualization, we could not assume that the two features are related.

d. Your discussion

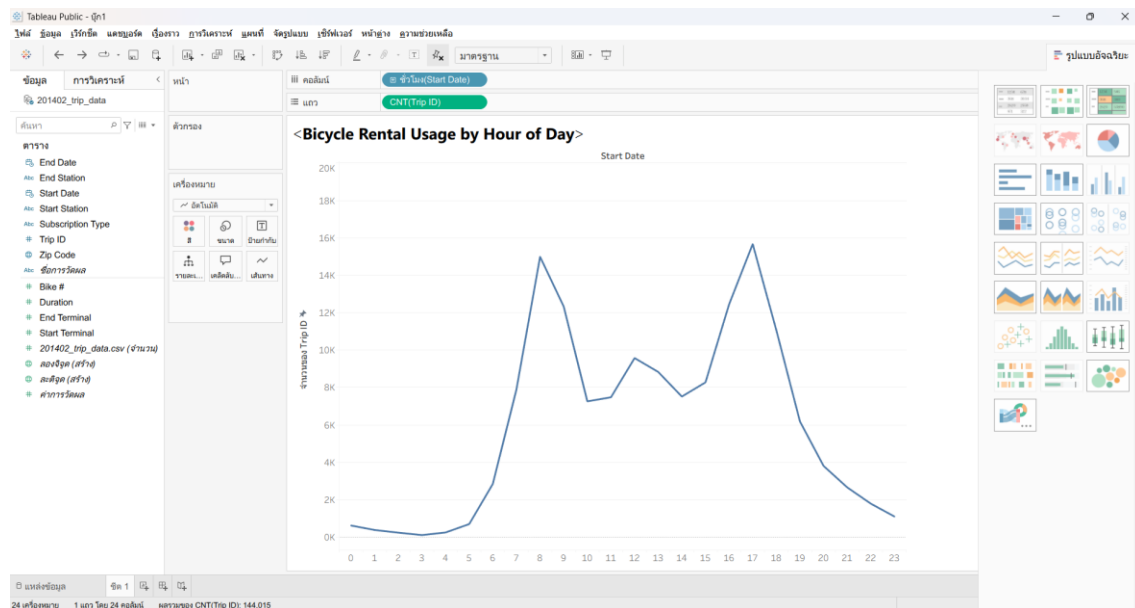
The plot did not really give any strong pattern. The least popular route has almost the same distance as the most popular route.

3. Use Tableau. Create more visualization explaining what happened to bicycle rental scheme. You may use additional dataset if needed.

a. Business question the visualization attempts to answer.

How was the bicycle rental scheme used throughout the day?

b. Visualization



c. Answer obtain (or not) from visualization

Yes, we can see that this people begin their trips at 3AM where their amount steadily increases up until 5AM. Then, from 5AM onwards; the number skyrocketed and peaked at 15k trips around 8AM. After that, bike sharing experienced sharp decline to around 7k-8k and fluctuated a bit until 16.00 hours where it suddenly reached the highest point at almost 16k by 6PM. Lastly, the number of trips steadily decreased as the night falls.

d. Your discussion

Well, the number of trips does relate to time-hour of the day as shown in this line graph. The number peaked during rush hours where people, everywhere, all at once, travel to-from work and their residence.